Sample Book

Ada Example

This small book is a test fixture. It has a title page numbered in roman numerals, four body pages numbered from one, a running header, and a page where the text is set in two columns. Nothing in it is meant to be read for its own sake, but every sentence is complete so that paragraphs end where a reader expects them to end.

Entropy

Entropy measures how much surprise a source of messages carries. A fair coin has one bit of entropy, because each toss is equally likely to land either way, while a coin that always lands heads has none, because nothing it says can surprise anyone.

Every message that a source can send carries some amount of information, and the content of one message is the logarithm of the inverse of its probability. Rare messages carry more of it, and the average over all messages is the entropy of the source. This is what makes entropy the natural measure of how far a message can be compressed.

A claim that is self-evident carries no surprise to those who already hold it, and so it adds no information for them. The same message can be news to one reader and noise to another, which is why entropy is always stated relative to what the receiver knows. In 1948 the idea was set out in full, and the figure 1996 below is the number of a later survey, not a year.

Joint entropy

Joint entropy is the entropy of two variables taken together. If the two are independent, it is the sum of their separate entropies. If one tells you something about the other, the joint entropy is smaller than that sum, and the difference is the mutual information between them.

Consider two coins tossed by the same hand. When the first lands heads the second is a little more likely to do so as well, since the hand repeats its own motion. The pair of results then carries less than two full bits, and a code for the pair can use fewer than two bits on average.

Conditional entropy is the entropy that remains in one variable once the other is known. It never exceeds the entropy of the variable on its own, and it equals that entropy only when the two variables are independent. A weather report that always agrees with the weather leaves no conditional entropy at all, while a report written at random leaves all of it.

The chain rule ties these quantities together. The joint entropy of two variables is the entropy of the first plus the conditional entropy of the second given the first. Applied again and again, the rule splits the entropy of a long message into the cost of each symbol given the symbols before it.

Students are told early in the course that a thing which is self-evident to a theorist may still need a proof, and the proof of the chain rule is short enough to give in full. It follows from the definition of conditional probability and the additivity of the logarithm, and it needs nothing else.

Mutual information is the difference between an entropy and a conditional entropy. It is symmetric, it is never negative, and it is zero exactly when the two variables are independent. Every later result in this book about what a channel can carry is a statement about this one quantity.

Alpha column opens the page with the first half of the text. Take a source that sends the letter A with probability one half and the letters B and C with probability one quarter each. Its entropy is one and a half bits, which is less than the logarithm of three that a source with three equally likely letters would have. A code that spends one bit on A and two bits on each of B and C reaches that figure exactly, and no code can do better on average.

Omega column holds the second half of the text. Now let two such sources run side by side without influencing one another. Their joint entropy is three bits, the sum of the two. Link them, so that the second repeats the first one time in four, and the joint entropy falls below three bits by exactly the information that each now carries about the other. The saving is real, and a joint code can collect it.

Channels

A channel takes the symbols a sender puts in and returns the symbols a receiver gets out. Noise makes the two differ. The capacity of a channel is the highest rate, in bits per symbol, at which messages can be sent so that the receiver recovers them with an error as small as one likes.

The noisy channel coding theorem says that every rate below the capacity can be reached and that no rate above it can. The proof is not constructive, and for decades it was not known how to build codes that come close to the limit. Modern codes do come close, and the gap between theory and practice is now small enough to measure in fractions of a decibel.

The theorem ends this short book. Entropy told us how much a source says, mutual information told us how much a channel passes on, and the capacity told us the most that any code could ever ask of it. Everything else is the engineering of getting there.